

The extended perturbation method: With applications to the New Keynesian model and the zero lower bound

Martin M. Andreasen

Grace Sun, Anastasia Chondrogianni

November 30, 2022

Contents

- 1 Introduction
- 2 The Extended Perturbation Method
- 3 A New Keynesian Model
- 4 The Zero Lower Bound in the New Keynesian Model
- 5 Conclusions

Introduction

Introduction I

- Background
 - The perturbation method: A profound impact on the DSGE literature
 - **Preserve nonlinearities**
 - **capture effects of uncertainty shocks**
 - **Determinants of risk premia**
 - The perturbation method is **computational efficient** - even for large model (10 states or more)

Introduction II

- **Drawback from the general perturbation methods:**
 - May be **unstable**
 - May not preserve **monotonicity and convexity** (Den Haan and De Wind (2012))
- A relatively low Taylor series expansion of the certainty equivalent component may generate unnecessary approximation errors
- **The second- and third-order perturbation approximations currently applied in the literature may not always be sufficiently accurate.**

Introduction III

- Contributions
 - **Improve accuracy** of the perturbation approximation by combining it with the **Extended Path** of Fair and Taylor (1983) .
 - Decomposing the policy function into (i) a deterministic component under certainty equivalence and (ii) a stochastic component containing the effects of uncertainty.
 - Certainty equivalent component $\Rightarrow$ Extended path;
 - Stochastic component $\Rightarrow$ Standard perturbation.
- Applications
 - A New Keynesian model with a new Zero Lower Bound Rule for the monetary policy rate: more **accurate** approximation of the model and **longer stays** at the (ZLB).

The Extended Perturbation Method

DSGE Models I

- The broad class of DSGE models

$$\mathbb{E}_t[\mathbf{f}(\mathbf{x}_t, \mathbf{x}_{t+1}, \mathbf{y}_t, \mathbf{y}_{t+1})] = \mathbf{0},$$

$$\mathbf{x}_t \equiv [\mathbf{x}'_{1,t} \ \mathbf{x}'_{2,t}]'$$

- Where $\mathbf{x}_{1,t}$ contains the endogenous state variables and $\mathbf{x}_{2,t}$ denotes the exogenous states, and $\mathbf{y}_t$ contains the control variables.
- The dynamics of $\mathbf{x}_{2,t}$ are given by:

$$\mathbf{x}_{2,t+1} = \mathbf{\Gamma}(\mathbf{x}_{2,t}) + \sigma \bar{\boldsymbol{\eta}} \boldsymbol{\epsilon}_{t+1},$$

- The exact solution to the considered class of DSGE models may then be expressed as:

$$\mathbf{y}_t = \mathbf{g}(\mathbf{x}_t, \sigma),$$

$$\mathbf{x}_{t+1} = \mathbf{h}(\mathbf{x}_t, \sigma) + \sigma \boldsymbol{\eta} \boldsymbol{\epsilon}_{t+1},$$

$$\boldsymbol{\eta} \equiv \begin{bmatrix} \mathbf{0}_{n_{x1} \times n_{\epsilon}} \\ \bar{\boldsymbol{\eta}} \end{bmatrix}.$$

DSGE Models II

- The perturbation parameter
 - $\sigma \geq 0$ scales the square root of the covariance matrix for the innovations η
 - To capture effects of uncertainty in the policy functions.
- Deterministic model under certainty equivalence

$$\mathbf{g}^{\text{CE}}(\mathbf{x}_t) \equiv \mathbf{g}(\mathbf{x}_t, \sigma = 0),$$

$$\mathbf{h}^{\text{CE}}(\mathbf{x}_t) \equiv \mathbf{h}(\mathbf{x}_t, \sigma = 0),$$

The extended perturbation method I

- **Key Finding:** The policy functions can be decomposed into two parts:
 - 1 A certainty equivalent solution
 - 2 A stochastic component

$$\mathbf{g}^{\text{stoch}}(\mathbf{x}_t, \sigma) \equiv \mathbf{g}(\mathbf{x}_t, \sigma) - \mathbf{g}^{\text{CE}}(\mathbf{x}_t),$$

$$\mathbf{h}^{\text{stoch}}(\mathbf{x}_t, \sigma) \equiv \mathbf{h}(\mathbf{x}_t, \sigma) - \mathbf{h}^{\text{CE}}(\mathbf{x}_t),$$

- The solution could be express as:

$$\mathbf{y}_t = \mathbf{g}^{\text{CE}}(\mathbf{x}_t) + \mathbf{g}^{\text{stoch}}(\mathbf{x}_t, \sigma),$$

$$\mathbf{x}_{t+1} = \mathbf{h}^{\text{CE}}(\mathbf{x}_t) + \mathbf{h}^{\text{stoch}}(\mathbf{x}_t, \sigma) + \sigma \boldsymbol{\eta} \boldsymbol{\epsilon}_{t+1}.$$

- 1 **extended path:** solves for $(\mathbf{g}^{\text{CE}}, \mathbf{h}^{\text{CE}})$ to any desired degree of accuracy by additional higher-order terms.

The extended perturbation method II

- An infinite Taylor series expansion of the certainty equivalent solution:

$$\mathbf{g}^{\text{CE}}(\mathbf{x}_t) = \sum_{k=0}^{\infty} \frac{\mathbf{g}(\mathbf{x}_{ss}, 0)_{\mathbf{x}^k}}{k!} (\mathbf{x}_t - \mathbf{x}_{ss})^{\otimes k},$$

$$(\mathbf{x}_t - \mathbf{x}_{ss})^{\otimes m} \equiv \underbrace{(\mathbf{x}_t - \mathbf{x}_{ss}) \otimes \dots \otimes (\mathbf{x}_t - \mathbf{x}_{ss})}_{m \text{ times}}$$

- 1 This procedure could be considered as a tractable way to “extend” the standard perturbation approximation by additional higher-order terms
- 2 This is not possible to include due to computational constraints in general perturbation method.
 $\Rightarrow$ Reducing approximation errors in the certainty equivalent part

The extended perturbation method III

Extended vs Standard perturbation

- ① For instance, the three-order approximation reads:

$$\begin{aligned} \mathbf{y}_t = & \sum_{m=0}^{\infty} \frac{\mathbf{g}(\mathbf{x}_{ss}, 0)_{\mathbf{x}^m}}{m!} (\mathbf{x}_t - \mathbf{x}_{ss})^{\otimes m} \\ & + \frac{1}{2} \mathbf{g}_{\sigma\sigma} + \frac{3}{6} \mathbf{g}_{\sigma\sigma\mathbf{x}} (\mathbf{x}_t - \mathbf{x}_{ss}) + \frac{1}{6} \mathbf{g}_{\sigma\sigma\sigma} \end{aligned}$$

- ② Subscripts to denote partial derivatives of a given function when evaluated at the steady state.
- ③ Compare to the standard perturbation method, the extended way add the

$$\sum_{m=4}^{\infty} \frac{\mathbf{g}(\mathbf{x}_{ss}, 0)_{\mathbf{x}^m}}{m!} (\mathbf{x}_t - \mathbf{x}_{ss})^{\otimes m}$$

to a standard third-order perturbation approx but not additional terms correcting for uncertainty.

The extended perturbation method IV

The conclusion of the extended perturbation method

- ① **Step 1:** Use standard perturbation to obtain all derivatives of $\mathbf{g}^{stoch}(\mathbf{x}_t, \sigma)$ and $\mathbf{h}^{stoch}(\mathbf{x}_t, \sigma)$ at the steady state to order m .
- ② **Step 2:** In any time period, use the extended path to compute $\mathbf{g}^{CE}(\mathbf{x}_t)$ and $\mathbf{h}^{CE}(\mathbf{x}_t)$ and approximate $\mathbf{g}^{stoch}(\mathbf{x}_t, \sigma)$ and $\mathbf{h}^{stoch}(\mathbf{x}_t, \sigma)$ by $\hat{\mathbf{g}}^{stoch}(\mathbf{x}_t, \sigma)$ and $\hat{\mathbf{h}}^{stoch}(\mathbf{x}_t, \sigma)$, respectively, using the derivatives from standard perturbation.

► [Jump to NK model](#)

A New Keynesian Model

A New Keynesian Model I

- Study of the accuracy and stability of standard and extended perturbation method, using a New Keynesian model with Calvo pricing.
- Why NK?
 - 1 First, New Keynesian models are the most popular DSGE models.
 - 2 Second, the **strong nonlinearities** imply that standard perturbation can generate explosive dynamics.

A New Keynesian Model II

- The representative household maximizes

$$U_t = E_t \sum_{l=0}^{\infty} \beta^l \left(\frac{c_{t+l}^{1-\phi_2}}{1-\phi_2} + \phi_0 \frac{(1-h_{t+l})^{1-\phi_1}}{1-\phi_1} \right) \quad (1)$$

where c_t is consumption and h_t is labor supply

- The household's real budget constraint is given by:

$$c_t + b_t + i_t = h_t w_t + r_t^k + \frac{R_{t-1} b_{t-1}}{\pi_t} + div_t \quad (2)$$

- Firms are perfectly competitive and produce final output y_t using intermediate goods y_i, t which are produced by monopolistic competitors
- Finally, monetary authority is specified by a standard Taylor Rule

$$\log(R_t/R_{ss}) = \rho \log(R_{t-1}/R_{ss}) + (1-\rho)(\kappa_\pi \hat{\pi}_t + \kappa_y \hat{y}_t) \quad (3)$$

A New Keynesian Model III

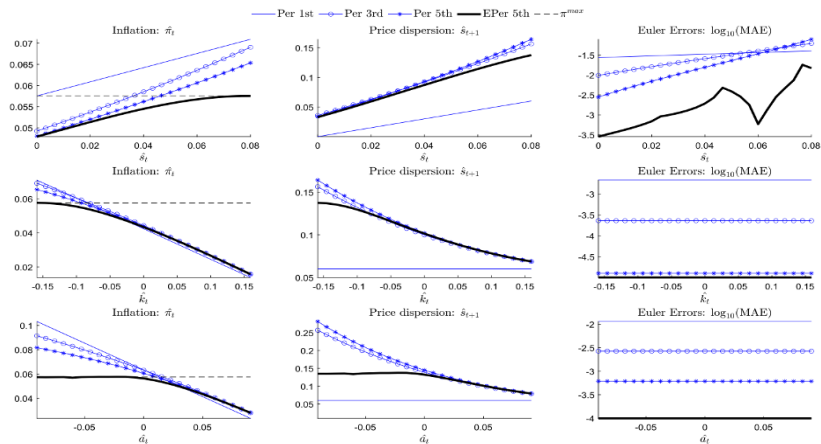
- There are two values for the steady-state inflation rate.
 - 1 Case A: $\pi_{ss} = 1.00$
 - 2 Case B: trend inflation $\pi_{ss} = 1.0005$

A New Keynesian Model IV: Comparison of two perturbation methods

		Standard Perturbation			Extended Perturbation		
		MAE	RMSE	MaxE	MAE	RMSE	MaxE
Case A	No trend inflation ($\pi_{ss} = 1.00$)						
	Panel I: On grid						
	1 st order	-2.24	-1.72	-0.55	-3.39	-2.63	-1.25
	2 nd order	-2.59	-2.03	-0.61	-3.42	-2.63	-1.23
	3 rd order	-2.82	-2.13	-0.52	-3.44	-2.63	-1.22
	4 th order	-3.04	-2.22	-0.52	-3.45	-2.63	-1.21
	5 th order	-3.20	-2.30	-0.53	-3.46	-2.63	-1.19
	Panel II: On simulated sample						
	1 st order	-2.84	-2.20	-0.77	-3.19	-2.23	-0.71
	2 nd order	-2.87	-1.98	-0.64	-3.21	-2.23	-0.71
	3 rd order	-2.59	-1.55	0.00	-3.23	-2.23	-0.56
	4 th order	-0.93	1.37	3.80	-3.24	-2.24	-0.56
	5 th order	0.07	2.22	4.62	-3.25	-2.24	-0.56
Case B	With trend inflation ($\pi_{ss} = 1.005$)						
	Panel III: On grid						
	1 st order	-2.10	-1.59	-0.45	-2.99	-2.29	-0.97
	2 nd order	-2.19	-1.49	0.22	-2.99	-2.29	-0.97
	3 rd order	-1.69	0.11	2.11	-2.99	-2.29	-0.96
	4 th order	-0.76	1.06	2.99	-2.98	-2.29	-0.95
	5 th order	-1.82	-0.59	1.18	-2.98	-2.29	-0.94
	Panel IV: On simulated sample						
	1 st order	-2.67	-2.11	-0.78	-2.90	-2.09	-0.59
	2 nd order	-2.58	-1.87	-0.63	-2.92	-2.10	-0.58
	3 rd order	-2.26	-1.41	0.16	-2.93	-2.10	-0.58
	4 th order	-0.22	2.21	4.66	-2.93	-2.11	-0.58
	5 th order	5.03	7.47	9.92	-2.94	-2.12	-0.58

A New Keynesian Model V: Stability Analysis II

Figure 1: Inflation and price dispersion at extreme state configurations



A New Keynesian Model V: Stability Analysis III

- **Price-Inflation Spiral**

- In **standard perturbation** at 3rd and 5th order inflation increases sharply for higher values of price dispersion $\hat{s}_t$. This increase leads to higher values of $\hat{s}_{t+1}$ in the next period, which in turn increases $\hat{\pi}_{t+1}$ and $\hat{s}_{t+2}$ (price-inflation spiral).
- **However**, inflation dynamics from **extended perturbation** are different since inflation **converges to a threshold**. (s_t increases more slowly and improves accuracy)

▶ [Jump to conclusion](#)

The Zero Lower Bound in the New Keynesian Model

The Zero Lower Bound in the New Keynesian Model

• The ideas:

- ① Following the financial crisis starting in 2008, an important requirement for the New Keynesian model is to respect the zero lower bound for the monetary policy rate. This is typically done by truncating the Taylor rule for $\log R_t$ at zero.
- ② However, such a shadow rate extension of the Taylor rule makes the New Keynesian model very demanding to solve because it introduces a **kink** in the model and several steady states.
- Formulate a Taylor rule for the net interest rate r_t instead of the gross interest rate $R_t = 1 + r_t$

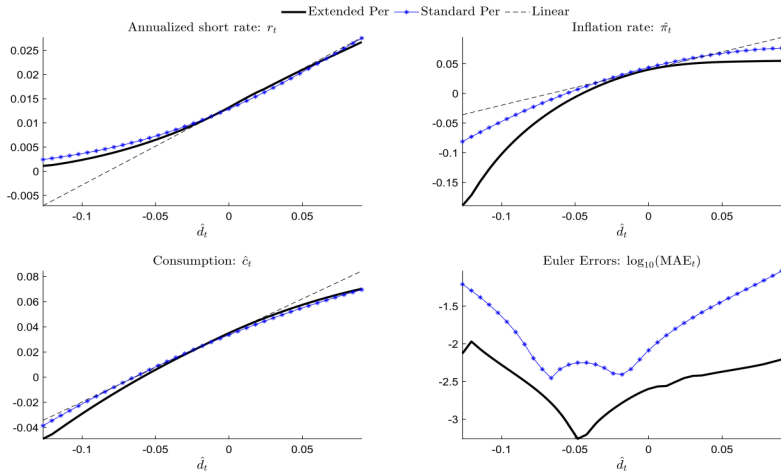
$$\log(r_t/r_{ss}) = \rho \log(r_{t-1}/r_{ss}) + \frac{1-\rho}{r_{ss}}(\kappa_\pi \hat{\pi}_t + \kappa_y \hat{y}_t),$$

previous rule:

$$\log(R_t/R_{ss}) = \rho \log(R_{t-1}/R_{ss}) + (1-\rho)(\kappa_\pi \hat{\pi}_t + \kappa_y \hat{y}_t)$$

The Zero Lower Bound in the New Keynesian Model

• Policy Function



Conclusions

Conclusions

- ① The explosive behavior of standard perturbation may be related to **inaccuracies in the certainty equivalent component** of the policy function.
- ② This paper introduces the extended perturbation approximation which calculates the **certainty equivalent more accurately** by adding all the higher-order terms.
- ③ The gain in accuracy **generates stable approximations of DSGE models** contrary to standard perturbation.
- ④ **Open floor: Any questions?**

Main References

- Den Haan, W. J. and De Wind, J. (2012). Nonlinear and stable perturbation-based approximations. *Journal of Economic Dynamics and Control*, 36(10):1477–1497.
- Fair, R. and Taylor, J. (1983). Solution and maximum likelihood estimation of dynamic nonlinear rational expectations models. *Econometrica*, 51(4):1169–85.

Thank you!

Appendix

Appendix A: Stability of the extended perturbation approximation

- **Certainty Equivalence part:** The imposed assumptions ensure stability under certainty equivalence, as all state variables (in the absence of shocks) return to the steady state when time tends to infinity.
- **Stochastic part:** Go beyond the third order, we cannot guarantee a stable approximation.

⇒ **Testing for stability**

- 1 unit root test .
- 2 simulate multiple sample paths and see if any of these simulations explode

⇒ **Inducing Stability**

- 1 if extended perturbation is unstable, then we suggest to use the modified state transition:

$$\mathbf{x}_{t+1} = \mathbf{h}^{\text{CE}}(\mathbf{x}_t) + \hat{\mathbf{h}}^{\text{stoch}}(\sigma) + (\hat{\mathbf{h}}^{\text{stoch}}(\mathbf{x}_t, \sigma) - \hat{\mathbf{h}}^{\text{stoch}}(\sigma))\Phi(\mathbf{x}_t) + \sigma\boldsymbol{\eta}\boldsymbol{\epsilon}_{t+1},$$

Appendix B: Computation of unit-free Euler errors

- Accuracy is evaluated by computing unit-free Euler errors at state x_t for all $n=16$ equations in the New Keynesian model.

1

$$MAE = \log_{10} \left(\sum_{t=1}^T \sum_{i=1}^{n=1} |E_{i,t}| / (nT) \right) \quad (4)$$

2

$$RMSE = \log_{10} \sqrt{\left(\sum_{t=1}^T \sum_{i=1}^{n=1} |E_{i,t}^2| / (nT) \right)} \quad (5)$$

3

$$\max E = \log_{10} \left(\max \left(\{ \{ E_{i,t} \}_{i=1}^n \}_{t=1}^T \right) \right) \quad (6)$$

Appendix C: Computational Issues I

Figure: Computational costs of extended perturbation

	Accuracy: MAE	Seconds	Average Iterations	Average N_t^*	Fraction by Extended Path
Panel A					
Start = 1st	-3.45	964	3.3	200	100%
Start = 4th	-3.45	881	2.4	200	100%
Panel B					
Start = 4th, $D_{ss} = 0.01$	-3.45	654	2.5	120	100%
Start = 4th, $D_{ss} = 0.03$	-3.45	333	2.5	72	100%
Start = 4th, $D_{ss} = 0.05$	-3.47	134	2.5	57	100%
Panel C					
Start = 4th, $D_{ss} = 0.05$, $\mathcal{R} = 10^{-4}$	-3.46	105	1.7	64	44%
Start = 4th, $D_{ss} = 0.05$, $\mathcal{R} = 10^{-3}$	-3.45	94	2.2	71	19%

- It is important to have an accurate terminal condition y_{t+N} and good starting values $\{x_{t+i}^{EP}, y_{t+i-1}^{EP}\}_{i=1}^N$
- Use of higher order derivatives for extended perturbation, except for when these elements are far from their steady-state (more than 50% or 100%).

Appendix C: Computational Issues II

Figure: Computational costs of extended perturbation

	Accuracy: MAE	Seconds	Average Iterations	Average N_t^*	Fraction by Extended Path
Panel A					
Start = 1st	-3.45	964	3.3	200	100%
Start = 4th	-3.45	881	2.4	200	100%
Panel B					
Start = 4th, $D_{ss} = 0.01$	-3.45	654	2.5	120	100%
Start = 4th, $D_{ss} = 0.03$	-3.45	333	2.5	72	100%
Start = 4th, $D_{ss} = 0.05$	-3.47	134	2.5	57	100%
Panel C					
Start = 4th, $D_{ss} = 0.05$, $\mathcal{R} = 10^{-4}$	-3.46	105	1.7	64	44%
Start = 4th, $D_{ss} = 0.05$, $\mathcal{R} = 10^{-3}$	-3.45	94	2.2	71	19%

- It is desirable to have a relatively short horizon N as the system grows linearly in N (usually $N_{min} = 50$, $N_{max} = 200$)
- It is better to select the shortest horizon where the largest element y_{t+N} within distance D_{ss} from the steady state.

Appendix C: Computational Issues III

Figure: Computational costs of extended perturbation

	Accuracy: MAE	Seconds	Average Iterations	Average N_t^*	Fraction by Extended Path
Panel A					
Start = 1st	-3.45	964	3.3	200	100%
Start = 4th	-3.45	881	2.4	200	100%
Panel B					
Start = 4th, $D_{ss} = 0.01$	-3.45	654	2.5	120	100%
Start = 4th, $D_{ss} = 0.03$	-3.45	333	2.5	72	100%
Start = 4th, $D_{ss} = 0.05$	-3.47	134	2.5	57	100%
Panel C					
Start = 4th, $D_{ss} = 0.05$, $\mathcal{R} = 10^{-4}$	-3.46	105	1.7	64	44%
Start = 4th, $D_{ss} = 0.05$, $\mathcal{R} = 10^{-3}$	-3.45	94	2.2	71	19%

- The unit-free Euler error are $r_t = f(x_t, x_{t+1}^{PE}, y_t^{PE}, y_{t+1}^{PE})$
- When $\max |r_t| > R$ where R denotes the tolerated errors in the c.e. component of the policy function we use extended perturbation.
- For this model $R = 10^{-4}$ or $R = 10^{-3}$ (typical range for New Keynesian models)

Appendix D: Estimation

- Due to modest computational costs of the extended perturbation it is possible to include the approximation in existing estimation routines for nonlinear DSGE models.
- Some estimators are:
 - 1 Simulated Method of Moments(SMM)
 - 2 Nonlinear Filtering Approach (requires few evaluations of the policy function)
 - 3 Bayesian Perspective
 - 1 Particle Filter (requires several evaluations of the policy function)
 - 2 Bayesian Inference